# Allaa Saboukh

AI Engineer

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LinkedIn

**Dynamic AI & Data Science Engineer** with 3 years of proven expertise in **Computer Vision, Deep Learning, and Advanced Analytics**. Skilled in building intelligent systems that transform complex data into actionable insights. Recognized for **technical excellence, innovation, and a results-driven mindset** that powers real-world AI impact.

## EXPERIENCE

**AI Engineer** • SANAD Professional Investment Company • Saudi Arabia (Remote) Apr '25 – Present

- Designed and deployed real-time **object detection and tracking models** (YOLO, MOT) for surveillance.
- Integrated AI solutions on **edge devices** (NVIDIA Jetson, OpenVINO) achieving faster real-time performance.
- Researched and implemented advanced **multi-object tracking** and scene understanding techniques.

**AI Engineer** • El7arefah • Egypt (Remote) Oct '24 – Feb '25

- Led AI performance optimization strategies, improving scalability and reliability of deployed models.
- Provided technical guidance on **model tuning, troubleshooting, and efficiency improvements**.

**AI/ML Engineer** • freelancer May '22 – Present

- Developed and fine-tuned **LLMs, GANs, VAEs, and Diffusion models** for diverse applications.
- Built NLP pipelines including **text classification, translation, and generative models**.
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- Delivered **custom AI solutions** that automated processes and improved client workflows

**AI Engineer** • Discover data (Freelancer) • Egypt (Remote) Oct '25 – Present
Predictive AI Framework for Nurse–Patient Interactions using NurViD + Digital Twin Simulation

- A comprehensive framework that unifies multilingual clinical text, video, and physiological data for predictive modeling.
- Integration of predictive AI with Virtual Shadowing and Digital Twin environments for healthcare applications.

## EDUCATION

**Kafrelsheikh University, Egypt**, kafer elshaikh, Egypt | Bachelor 2020 – 2024
Artificial Intelligence

**National Telecommunication Institute (NTI), Alexandria, Egypt**, Alexandria, Egypt 2024 – 2025
data analysis and visualization

## PROJECTS

**COVID-19 Detection using Convolutional Neural Networks (CNNs)**

- Developed a deep learning model using Convolutional Neural Networks (CNNs) to detect and classify covid-19 from medical images accurately.

**Early Warning System (EWS)**

- Work has been done on the part of the brain to detect diseases and emergency issues with computer vision specially GANs model and is currently working on the rest of the body to create an integrated system.

**Meal planning**

- AI system predicting meal impact on blood sugar using CV & ML.

**diabetic retinopathy detection**

- Deep learning pipeline (ResNeXt101, Inception-V3) with >93% accuracy.

**Food Recognition for Calorie Estimate**

- YOLOv5-based detection system integrated with nutrition database.

## VOLUNTEERING

**GDSC** | Alazhar, cairo 2022 – 2023
Instructor
Taught machine learning concepts, guided hands-on projects, and mentored students in Python, TensorFlow, and model optimization. Delivered workshops and personalized coaching.

**IEEE** | Kafrelsheikh 2021 – 2022
trainee
Gained hands-on experience in computer vision techniques, image processing, and deep learning models. Worked on real-world projects and collaborated with peers to enhance practical skills.

## TEChHNICAL SKILL

**Programming Languages:** Python, C/C++, JavaScript
**Deep Learning Frameworks:** PyTorch, TensorFlow, Keras, FastAI
**Computer Vision:** OpenCV, YOLOv5/v8, MediaPipe, Detectron2
**Data Science & ML:** Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
**Deployment & APIs:** Flask, FastAPI, Streamlit, Docker
**Databases:** SQL, MongoDB, Microsoft SQL Server
**Tools & Platforms:** Google Colab, Jupyter, Git, Linux, AWS, Google Cloud
**Others:** Model Optimization, Edge Deployment, Data Preprocessing, Feature Engineering

## LANGUAGE

**Arabic** (native)
**English** (B2+)
**German** (A2)